

Sanika S. Khadkikar

Fourth Year Graduate Student at The Pennsylvania State University

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EDUCATION

The Pennsylvania State University, State College, PA <i>Ph.D. in Physics</i> <i>Exploring the fundamental physics of neutron stars using astrophysics and gravitational waves</i> Advisor: Bangalore Sathyaprakash	2022 -
Birla Institute of Science and Technology, Pilani, India <i>M. Sc. (Hons) in Physics</i> <i>B.E. (Hons) in Mechanical Engineering</i> <i>Quasi-stationary sequences of hyper-massive neutron stars with exotic equations of state</i> Advisor: Sarmistha Banik	2017 - 2022
The Pennsylvania State University, State College, PA <i>BITS Pilani exchange program</i> <i>Binary neutron star post-merger signal analysis using wavelet transforms</i> Advisor: Bangalore Sathyaprakash and Sujith R.	2021 - 2022

RESEARCH INTERESTS

Neutron stars, gravitational wave data analysis, black holes, multimessenger astronomy, compact object binaries, nuclear astrophysics, next generation gravitational wave detectors, dark matter

SELECTED FELLOWSHIPS AND HONORS

LIGO Compact Binary Coalescence Award <i>Laser Interferometer Gravitational-wave Observatory (LIGO) Collaboration</i> <i>Recognition for outstanding contribution towards collaboration science</i>	2026
Peter Clay Eklund Award for Scientific Communication <i>Pennsylvania State University</i> <i>Awarded by the Department of Physics for excellence in scientific communication of research</i>	2026
Graduate Student International Research Award <i>Pennsylvania State University</i> <i>Sole recipient selected from a university-wide pool of over 12,600 graduate students across all Penn State campuses for excellence in international research.</i>	2026
John Randall Shuman Troxell Memorial Scholarship in Physics <i>Pennsylvania State University</i> <i>Annual merit-based fellowship awarded by the Eberly College of Science at Penn State</i>	2025
Edward M. Frymoyer Honors Scholarship <i>Eberly College of Science, Pennsylvania State University</i> <i>Awarded for an excellent academic record and achievements</i>	2025
Edward A. and Rosemary A. Mebus Graduate Fellowship in Physics <i>Eberly College of Science, Pennsylvania State University</i> <i>Awarded for an excellent academic record and achievements</i>	2025
Division of Gravitational Physics Best Student Speaker <i>American Physical Society</i> <i>Awarded for the best student presentation at the annual APS Eastern Gravity meeting</i>	2025

Physics and Astronomy for Women+ Research Travel Grant <i>Pennsylvania State University</i> <i>Merit-based scholarship awarded for women+ physicists to travel to a conference</i>	2025
Rick Robinett Service Award <i>Pennsylvania State University</i> <i>Awarded by the Department of Physics for efforts in science communication and outreach to a larger community</i>	2025
Distinguished Student Award <i>American Physical Society</i> <i>Awarded by APS's Forum on International Physics to outstanding students in any field of physics</i>	2025
W. Donald Miller Graduate Fellowship <i>Pennsylvania State University</i> <i>Annual merit-based fellowship awarded by the Eberly College of Science at Penn State</i>	2025
David C. Duncan Graduate Fellowship <i>Pennsylvania State University</i> <i>Annual merit-based fellowship awarded by the Eberly College of Science at Penn State</i>	2025
Division of Gravitational Physics Travel Grant <i>American Physical Society</i> <i>Travel grant awarded by APS's Division of Gravitational Physics to allow student members to present their work at the APS meeting.</i>	2025
Peter Clay Eklund Award for Scientific Communication (Honorable mention) <i>Pennsylvania State University</i> <i>Awarded by the Department of Physics for excellence in scientific communication of research</i>	2025
ACCESS Computing Grant (Co-PI) <i>National Science Foundation</i> <i>500k Compute Hours awarded by the National Science Foundation as a Co-PI</i>	2024
I am STEM Award <i>Pennsylvania State University</i> <i>Student speaking contest to inspire future scientists in STEM</i>	2024
Homer F. Braddock Scholarship in Biology, Chemistry, and Physics <i>Pennsylvania State University</i> <i>Merit-based scholarship awarded by the Eberly college of Science at Penn State to first-year graduate students</i>	2022
Off-Campus International Master's Thesis Fellowship <i>Birla Institute of Technology and Science</i> <i>Merit-based scholarship awarded by BITS-Pilani to complete an exchange year in a foreign country</i>	2021
Charpak Indo-France Research Scholarship <i>Government of France</i> <i>Ranked first nationwide in the highly competitive Charpak Research Fellowship program, awarded to fewer than 1% of applicants from India to intern in France.</i>	2021
LIGO Summer Undergraduate Research Fellowship (SURF) <i>California Institute of Technology</i> <i>Selected as one of four students in India for a fully funded research internship at the LIGO Laboratory, Caltech.</i>	2020

BITS Pilani Merit Scholarship <i>Birla Institute of Technology and Science</i> <i>Awarded 50 % tuition waiver for four consecutive years</i>	2018–2021
INSPIRE Award <i>Government of India</i> <i>One of 25 high school students across India to receive an award from the Government of India for scientific aptitude and research potentials</i>	2014

INVITED TALKS

Department Colloquium at University of California, San Diego <i>San Diego, CA</i>	<i>Upcoming</i>
Seminar at University of Southampton <i>Southampton, UK</i>	2026
Seminar at Royal Holloway University of London <i>Egham, UK</i>	2026
Seminar at University of Portsmouth <i>Portsmouth, UK</i>	2026
Seminar at University of Birmingham <i>Birmingham, UK</i>	2026
Seminar at CNRS Strasbourg <i>Strasbourg, France</i>	2026
Seminar at Sapienza University of Rome <i>Rome, Italy</i>	2026
Seminar at Gran Sasso Science Institute <i>L'Aquila, Italy</i>	2026
Seminar at Utrecht University <i>Utrecht, Netherlands</i>	2026
Seminar at Friedrich Schiller University Jena <i>Jena, Germany</i>	2026
Seminar at University of Potsdam <i>Potsdam, Germany</i>	2026
Seminar at Max Planck Institute for Gravitational Physics <i>Potsdam, Germany</i>	2026
Seminar at DESY Zeuthen <i>Zeuthen, Germany</i>	2026
Invited talk at APS 2026 Global Summit <i>Denver, CO</i>	2026
Seminar at Indian Institute of Technology, Bombay <i>Mumbai, India</i>	2025
Lecture at State College Area High School <i>State College, PA</i>	2025
Seminar at LIGO Hanford Observatory <i>Hanford, WA</i>	2025
Lecture at the Institute of Nuclear Theory <i>Seattle, WA</i>	2025
Astronomy on Tap <i>State College, PA</i>	2025

AstroChat Talk at University of California, Berkeley <i>Berkeley, CA</i>	2025
Department Colloquium at California State University Fullerton <i>Fullerton, CA</i>	2025
Einstein Telescope Nuclear Astrophysics Call <i>Virtual</i>	2025

CONTRIBUTED TALKS

Prospects in Theoretical Physics Summer School (Poster Presentation) <i>Princeton University, NJ</i>	2025
Nuclear Physics from Multi-Messenger Mergers Summer School <i>Indiana University, IN</i>	2025
American Physical Society Eastern Gravity Meeting <i>University of New Hampshire, NH</i>	2025
Neighborhood Workshop <i>State College, PA</i>	2025
American Physical Society Global Summit Meeting 2025 <i>Anaheim, CA</i>	2025
American Astronomical Society 245 Meeting <i>Oxon Hill, MD</i>	2025
LIGO-Virgo Collaboration Meeting <i>virtual</i>	2024
American Physical Society April Meeting 2024 <i>Sacramento, CA</i>	2024
Penn State Primordial Universe and Gravity Seminar <i>State College, PA</i>	2021, 2023, 2024, 2025
LIGO-SURF Presentation <i>California Institute of Technology (virtual)</i>	2020
Finesse Interferometric Modelling Workshop and Hackathon <i>IUCAA, Pune, India</i>	2020

PUBLICATIONS

I have authored 15 papers with > 450 citations, including 6 short-author papers with > 280 citations, combined, as of October 30, 2025.

SHORT-AUTHOR PUBLICATIONS

- **Khadkikar, S.**, & Singh, D. (2025). A case study of GW190425 for classifying binary neutron star versus binary black hole mergers and constraining asymmetric dark matter with gravitational wave detectors. arXiv. <https://arxiv.org/abs/2507.07895>
- **Khadkikar, S.**, Gupta, I., Kashyap, R., Chandra, K., Gamba, R., et al. (2025, February 5). Precise and accurate neutron star radius measurements with next generation gravitational wave detectors. Physical Review D, 112(6), 063020. <https://doi.org/10.1103/PhysRevD.112.063020>
- Gupta, I., et al. incl **Khadkikar, S.**, (2023). Characterizing gravitational wave detector networks: From A# to Cosmic Explorer. Class.Quant.Grav. 41 (2024) 24, 245001. <https://doi.org/10.1088/1361-6382/ad7b99>
- Evans, M., et al. incl **Khadkikar, S.**, (2023). Cosmic Explorer: A submission to the NSF MPSAC ngGW Subcommittee. arXiv. <https://arxiv.org/abs/2306.13745>

- **Khadkikar, S.**, Mangat, C. S., and Banik, S. (2022). Quasi-stationary sequences of hyper-massive neutron stars with exotic equations of state. *Journal of Astrophysics and Astronomy*, 43(2), 57. <https://doi.org/10.1007/s12036-022-09849-0>
- **Khadkikar, S.**, Raduta, A. R., Oertel, M., and Sedrakian, A. (2021). Maximum mass of compact stars from gravitational wave events with finite-temperature equations of state. *Physical Review C*, 103(5), 055811. <https://doi.org/10.1103/PhysRevC.103.055811>

COLLABORATION PUBLICATIONS

Member of the LIGO–Virgo–KAGRA (LVK) Collaboration since 2020. Served about eight distributed weeks during O4 on the Parameter Estimation Rotation (PE ROTA) and currently act as liaison for the GWTC-5 study as one of four analysts responsible for production runs and validation across multiple Bayesian inference pipelines.

- LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, A. G. Abac et al. incl **Khadkikar, S.**, (2025, September 9). *Directed searches for gravitational waves from ultralight vector boson clouds around merger remnant and galactic black holes during the first part of the fourth LIGO Virgo KAGRA observing run.* arXiv: <https://arxiv.org/abs/2509.07352>
- KAGRA Collaboration, Virgo Collaboration, and LIGO Scientific Collaboration, A. G. Abac et al. incl **Khadkikar, S.**, (2025, September 9). *GW250114: Testing Hawking’s area law and the Kerr nature of black holes.* *Physical Review Letters*, 135(11), 111403. <https://doi.org/10.1103/PhysRevLett.135.111403>
- LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, A. G. Abac et al. incl **Khadkikar, S.**, (2025, August 28). *Upper limits on the isotropic gravitational wave background from the first part of LIGO Virgo and KAGRA’s fourth observing run.* arXiv: <https://arxiv.org/abs/2508.20721>
- LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, A. G. Abac et al. incl **Khadkikar, S.**, (2025, August 25). *Open data from LIGO Virgo and KAGRA through the first part of the fourth observing run.* arXiv: <https://arxiv.org/abs/2508.18079>
- LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, A. G. Abac et al. incl **Khadkikar, S.**, (2025, August 25). *GWTC 4.0: Updating the gravitational wave transient catalog with observations from the first part of the fourth LIGO Virgo KAGRA observing run.* arXiv: <https://arxiv.org/abs/2508.18082>
- LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, A. G. Abac et al. incl **Khadkikar, S.**, (2025, July 16). *All sky search for long duration gravitational wave transients in the first part of the fourth LIGO Virgo KAGRA observing run.* arXiv: <https://arxiv.org/abs/2507.12282>
- LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, A. G. Abac et al. incl **Khadkikar, S.**, (2025, July 16). *All sky search for short gravitational wave bursts in the first part of the fourth LIGO Virgo KAGRA observing run.* arXiv: <https://arxiv.org/abs/2507.12374>
- LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, A. G. Abac et al. incl **Khadkikar, S.**, (2025, July 10). *GW231123: A binary black hole merger with total mass 190–265 solar masses.* arXiv: <https://arxiv.org/abs/2507.08219>

TO BE SUBMITTED SOON

- **S. Khadkikar** and B. S. Sathyaprakash. “A Fully Geometric Approach to Model Selection in Data Analysis”. *Expected In: (2025)*.
- **S. Khadkikar**. “The Fault Is Not in Our Stars, but in Our Priors: Physics-Informed Priors for Neutron Star Inference”. *Expected In: (2025)*.
- L. Suleiman, **S. Khadkikar**, and S. Guillot. “Systematic Biases in NICER Neutron Star Radius Measurements”. *Expected In: (2025)*.
- **S. Khadkikar**, M. Emma, A. Abac, N. Kunert, T. Dietrich, and V. Sagun. “Can Dark Matter Halos Around Neutron Stars Have Detectable Signatures in Gravitational Waves?”. *Expected In: (2025)*.

TEACHING AND MENTORSHIP

Graduate Teaching Assistant <i>Department of Physics, Pennsylvania State University</i> Quantum Information and Computing Introductory Electromagnetism Introductory Mechanics	2022–2024
Undergraduate Teaching Assistant <i>Department of Physics, BITS Pilani</i> Statistical Mechanics	2020

LEADERSHIP AND ACADEMIC SERVICE

Student Representative <i>Executive Committee of the American Physical Society's Division of Gravitational Physics</i> Elected Student Representative to the APS Division of Gravitational Physics Executive Committee following nomination by the DGRAV Nominating Committee and an open election of the division membership.	2026-2028
Council Member <i>Gravitational-Wave Early Career Scientists (GWECS)</i> Represented the Cosmic Explorer Consortium on the GWECS council and contributed to efforts supporting early-career researchers through resources, opportunities, and community initiatives.	2025
Chair of the Early Career Scientists Group <i>Cosmic Explorer Consortium</i> Coordinated early-career activities, mentorship efforts, and liaison with the executive board.	2025
GAPP Graduate Student Liaison <i>Department of Physics, Pennsylvania State University</i> Arranged and managed the Gravity, Astrophysics and Particle Physics seminars with faculty.	2024
Gravi-tea Time Podcast (Co-creator) <i>Institute for Gravitation and the Cosmos, Penn State</i> Produced episodes explaining breakthroughs in astrophysics for advanced undergrads and grads.	2024
Founder, Physics Journal Club <i>BITS Hyderabad</i> Fostered collaborative learning and research discussions.	2021
Secretary, AD ASTRA Astronomy Club <i>BITS Hyderabad</i> Initiated projects (e.g., horn antenna for galaxy rotation curves), organized fests and trivia sessions.	2019
Team Manager & Suspension Engineer, Vulcan Racing <i>BITS Hyderabad</i> Managed off-roading automobile team; contributed to suspension design and project coordination.	2019

OUTREACH AND SOCIAL SERVICE

Host, Astro[sound]bites Podcast <i>American Astronomical Society</i> One of four hosts that host this astronomy podcast for general outreach.	2025
Author, GW250114 Factsheet <i>LIGO Education and Public Outreach (EPO) Team</i> Designed the official GW250114 factsheet, translating key scientific results for broad public and media engagement.	2025
Outreach Chair of PAW+ <i>Physics and Astronomy for Women+, Pennsylvania State University</i> • Organized invited physics talks from renowned physicists for women and other underrepresented groups in physics. • Organized a fully physics-based experiment section for the Easterly Parkway School Fair in State College for children from kindergarten to grade 12. • Designed and coordinated a physics-themed escape room for the Haunted-U event at Penn State near Halloween to engage young minds in science.	2025
Graduate Student Newsletter Editorial Team <i>Eberly College of Science, Pennsylvania State University</i> • Co-editor for the Graduate Student Newsletter circulated throughout Penn State. • Interviewed several individuals to highlight their research in the Graduate Student Spotlight section. • Helped coordinate the publication of LIGO-related press releases through the official Penn State communications office.	2025
PAW Pals Volunteer <i>Physics and Astronomy for Women+, Pennsylvania State University</i> Led demonstrations and discussions in local elementary schools to engage kids in science.	2023–2025

PROFESSIONAL REFERENCES

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